



MB 580: AI for Finance

Course Code	MB 580
Course Name	AI for Finance
L-T-P-C	2-0-0-2
Intended for	MBA
Mutual Exclusion	NA
Approval	52nd BoA (Nov 2023)

Course Contents

- **Introduction to AI in Finance:** Overview of AI and Machine Learning Applications in Finance; Big Data and Cloud Computing in Financial Analysis; Fundamental Machine Learning Techniques: Regression, Classification; Ethical Implications of AI in Financial Services. (4 Hours)
- **Data Collection and Preparation:** Financial Data Sources: Market, Transactional, and Alternative Data; Preprocessing Financial Data: Cleaning and Normalization; Feature Engineering and Dimensionality Reduction Techniques; Time-Series Data Management. (6 Hours)
- **Predictive Modeling and Algorithmic Trading:** Building Predictive Models for Market Forecasting; Supervised and Unsupervised Learning Applications; Trading Algorithms and High-Frequency Trading; Sentiment Analysis of Financial News and Social Media. (6 hours)
- **Risk Management and Fraud Detection:** AI Techniques in Credit Risk Modeling; Fraud Detection Algorithms: Outlier Detection and Anomaly Detection; Portfolio Risk Management with Machine Learning; Stress Testing and Scenario Analysis. (6 hours)
- **Advanced Topics and Emerging Trends:** Deep Learning Models: Neural Networks, LSTM; Reinforcement Learning for Portfolio Optimization; Natural Language Processing in Financial Document Analysis; Regulatory Compliance with AI: Regtech and Explainability. (6 hours)

Text Books

1. NA

References

1. NA